L4
Intermediate loops

Join Discord - https://tinyurl.com/ly-discord

Type Casting

Implicit → Compiler will convert automatically

```
int a = 4;
double b = a;
```

4

4.0

```
int c = 4;
long d = c;
```

$\boxed{4}$

↓ long

long

int

int → $-2 * 10^9$ to $2 * 10^9$ (smaller) long → $-2 * 10^{18}$ to $2 * 10^{18}$ (big)

small → big

values

10^{18}

~~int~~

long a = 4;

int b = (int) a;

big → small

Force Convert → loss of data

big → small

loss of data

Explicit type casting

```
double c = 8.23;
float d = c;
```

Error

(float)

```
float d = (float) c;
```

Type casting

int a = 1000;

→ integer

long b = 1000;

→ integer

long b = 1000 000 000 000 L/l

→

By default any integer value is treated as int double a = 5.0;

Decimal ~ double;

big to small

float b = 4.5;

float b = (float) 4.5;

float

decimal

float b = 4.5f;

float

while → digits

for → start - - - - - - - end

for(initialization ; test condition ; update expression)

```
for( int i = 1; i <= 3; i = i + 1)
{
  S.O.Pln (i);
}
```

1 — int i = 1

2 — i <= 3

3 — S.O.Pln (i);

4 — i = i + 1

1
2
3

i = ~~1~~ ~~2~~ ~~3~~ 4

i = 4, 4 <= 3

↓

false → STOP

```
for ( int i = 1;  i <= 5;  i = i + 3)
{
    S.O.Pln(i);
}
```

$i = i + 3 \rightarrow 4$

$i = 7$

true

step I $i = 1$, $1 <= 5$, print 1, $i = 4$

step II $i = 4$, $4 <= 5$, print 4, $i = 7$

III $i = 7$, $7 <= 5$ → false, STOP

1
4

Print all numbers from 1 to n.

↳ User input

n → input

1 to n → for loop

Print all odd numbers from 1 to n

while → digits

1234

4, 3, 2, 1

intialization → int i = 1; → 1

while (test condition) → while (i <= 3) → 2

{ → { → 3

S.O.Pln (i);

update statement → i = i + 1; → 4

} → }

1
2
3

true,

I, i = 1, 1 <= 3, i = i+1 = 2

II, i = 2, 2 <= 3, i = i+1 = 3

III, i = 3, 3 <= 3, i = i+1 = 4 → stop

457

$n/10 \rightarrow \frac{457}{10} = 45$

Find last Digit $\rightarrow$ $n \% 10$ = last digit

Remove last digit

$$\begin{array}{r|l|l} 10 & 457 & 45 \\ & -450 & \\ \hline & 7 & \end{array}$$

7 $\rightarrow$ Remainder

$n = n/10;$

$n = n/10$

$n = 457$

7 5 4

reverse digits

```
while ( n > 0 )
{
    int lastDigit = n%10;
    S.O.Pln(lastDigit);
    n = n/10;
}
```

lD = 7 7

n = 45

lD = 5 5

n = 4

lD = 4 4

n = 4